

S&P Sentinel

Real-time financial-text risk signals driving index rebalancing and portfolio stress testing

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S&P Global & CRISIL Campus Hackathon 2026 | Individual submission

Built and evaluated on real public data: 4,117 + 2,388 held-out labeled financial tweets, 2,518 real news articles, 2,617 market-reaction events, 13.3M card transactions, live GDELT and SEC EDGAR feeds.

Problem & approach

Turn unstructured news and social text into machine-readable risk signals, then act on them

- **Problem:** material events (defaults, rate moves, sanctions, cyber incidents) surface first in text; banks need structured signals fast and must trust them.
- **Engine output per company:** sentiment (-1..1), event class, impact (1-10), evidence spans, action eligibility; served by API, SSE and a JSONL file.
- **Trust:** every model is trained and scored on real held-out data; metrics in this deck are read from docs/metrics.json (scripts/run_evaluation.py) and data facts from the committed data files.
- **Module A** (tactical): a 20-name mock S&P 100 index re-weighted by sentiment, with caps and a weights-over-time dashboard.
- **Module B** (strategic): event-driven stress tests on a \$610M funded book (wholesale + retail credit from real transactions + equities), shocks calibrated on five real crisis windows.
- **Real time:** a recorder captures GDELT (15-min) and SEC 8-K filings; the app itself makes no network calls.

System design & data flow

S&P SENTINEL — SYSTEM ARCHITECTURE & DATA FLOW

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OFFLINE APP RUNTIME
LIVE FEEDS VIA RECORDER

1. REAL DATA SOURCES

Downloaded, checksummed, licensed

- GDELT 2.0 GKG 15-min (live recorder)
- SEC EDGAR 8-K filings (live recorder)
- Polygon 2023 news: 5,548 articles
- Kaggle stock tweets: 862k (2017-18)
- HF labeled fin. tweets: 21k + 12k
- Kaggle card txns: 13.3M -> Module B
- yfinance closes; SHA-256 manifest

2. REPLAY, DEDUP & RELEVANCE

Badged sources, offline runtime

- Source registry: synthetic / historical / live
- Logical replay clock, live append
- Exact hash + MinHash-LSH near-dup
- Lead-sentence index (headline reposts)
- 24 h window; 100k tweets in 26 s
- Social spam & relevance filter
- No network calls in the app runtime

3. NLP RISK ENGINE

Trained and validated on real data

- Entity linker, S&P 500 universe: P 0.971
- One signal per company (multi-entity)
- Event classifier: macro-F1 0.846
- (4,117 held-out real tweets)
- Calibrated abstention + evidence gates
- precision 0.932 when it fires
- Sentiment: macro-F1 0.778 (lexicon 0.469)
- Rate cut / hike direction rules
- Impact 1-10: market-calibrated decile
- vs abnormal return: rho 0.151
- Rubric components + evidence spans
- 8-K cyber / credit / routine filings
- p95 0.015 s per record on CPU

4. MODULE A: INDEX REBALANCER

Sentiment-tilted mock S&P 100 index

- 20 names, equal base weights
- Time-decayed sentiment EMA per name
- $w = \text{base} \times (1 + k \times \text{EMA})$, long-only
- Caps: 20% per name, 40% per sector
- Weights over time + reasons + turnover
- Back-test on real tweets & prices
- Reported honestly: no proven alpha

5. MODULE B: STRESS ENGINE

\$610M funded + \$150M swap notional

- Wholesale \$500M (synthetic book)
- Retail/SME \$50M from real card txns
- Equity \$60M, betas from real prices
- ECL, duration, signed DV01, beta
- Shocks from 5 real crisis windows
- Contagion graph; CYBER fraud op-loss
- Reconciled; NO_EXPOSURE flagged

6. API, AUDIT & TERMINAL UI

Localhost FastAPI + React

- POST /api/analyze, /api/replay/*
- GET /api/signals, SSE event stream
- data/signals.jsonl file sink (PS)
- GET /api/index/* (Module A)
- POST /api/stress/scenario, /custom
- SQLite audit store, JSON/CSV export
- React terminal: live feed + inspector
- Module A weights-over-time chart
- Module B before/after + waterfall
- Real-data metrics panel
- docs/metrics.json: single source
- CI recomputes real-data suites

Implementation highlights

Key modules, technology choices and why

Component	What it does	Why this choice
Entity linker	S&P 500 universe from SEC/Wikipedia; spans equal the text; one signal per company. P 0.971 / R 0.588.	Rule-based and auditable; role guards stop rating agencies and brokers being linked as the subject.
Event classifier	TF-IDF + calibrated logistic regression, 8 PS classes + CYBER/SUPPLY_CHAIN; abstention threshold chosen on a dev split.	CPU-fast, explainable; trained on real labeled tweets and real 8-K filings.
Sentiment	Real-data TF-IDF model (macro-F1 0.778); policy rules make rate cuts positive and hikes negative.	Outperformed off-the-shelf FinBERT in a prototype comparison; FinBERT stays an optional tier.
Impact 1-10	Gradient-boosted model of next-day abnormal return, mapped to deciles; rubric kept for explanation.	Validated against what markets actually did, not hand-picked weights.
Dedup + relevance	Exact hash + MinHash-LSH with a lead-sentence index and a 24 h window; spam / no-context filter for social.	Bounded, near-linear index instead of an $O(n^2)$ scan; syndicated copies never re-trigger stress.
Module B engine	Loan ECL, bond duration, swap DV01, equity beta; contagion graph; CYBER loss from observed fraud rates; reconciliation invariant.	Shocks measured from real windows; funded value and derivative MTM separated.

Key results (real held-out data)

Metric	Measured	Baseline / target	Scenario (measured market move)	Portfolio P&L	% of funded
Event macro-F1 (4,117 tweets)	0.846	keyword 0.286, old 0.060; target 0.70	COVID-19 liquidity shock (Feb-Mar 2020)	-\$96.7M	-15.9%
Event precision when it fires	0.932	fires on 25% (abstains otherwise)	Fed 75 bp surprise hike (Jun 2022)	-\$14.7M	-2.4%
Sentiment macro-F1 (2,388 tweets)	0.778	lexicon 0.469; target 0.75	Fed pivot rally (Nov-Dec 2023)	+\$13.8M	2.3%
Entity linking (2,518 articles)	P 0.971 / R 0.588	target P 0.90	Russia-Ukraine invasion (Feb-Mar 2022)	-\$5.9M	-1.0%
Impact vs abnormal return (2,617)	rho 0.151	95% CI [0.113, 0.190]; rubric 0.098	SVB failure (Mar 2023)	-\$11.4M	-1.9%
Large moves, top vs bottom decile	30.1% vs 2.3%	abnormal z > 2	PS example: equities -10%, rates +200 bp	-\$23.3M	-3.8%
Routine 8-Ks sent to stress path	14.0%	141 held-out filings	Module A back-test (443 days, k=1.5): active return +0.12%/yr, tracking error 0.47%, t = 0.35: sentiment reweighting works, no proven alpha.		
Latency p95 per record (CPU)	0.015 s	96 records/s; target 2 s			

Domain impact

Why it matters for a credit-risk and ratings business

- **Flight to quality, measured:** in the SVB, COVID and Russia windows the 10-year Treasury yield *fell* while spreads widened; only Fed tightening raised it. The shock catalog encodes this instead of assuming every crisis lifts yields.
- **Credit triangle:** the SVB window's +139 bp HY spread implies +2.3 pp PD at 60% LGD, independently supporting the +150 bp / +2.5 pp single-name credit shock.
- **Retail credit from real transactions:** 2,000 borrowers and 13,305,915 card transactions pooled into sector x rating tranches; cyber losses use observed fraud rates by sector.
- **Impact you can defend:** the 1-10 score predicts the size of the next-day abnormal move out of sample, so analysts can rank events by expected market impact.
- **Auditability:** evidence spans, model versions, block reasons and a reconciliation invariant on every stress run; one JSON file holds every metric.

Limitations & next steps

- Headline NLP metrics are on financial *tweets*; agreement with LLM silver labels on news is lower (sentiment 0.47).
- Reaction-phrased headlines ('shares slide after...') are often abstained.
- Severity MAE vs human labels is pending: a 300-row real-news set is sampled (data/eval/holdout_v2_news.todo.csv) and must be labeled by hand.
- Simplified valuation: single-period ECL, no convexity, linear DV01; wholesale book and equity sizes are synthetic; retail sleeve is scaled to \$50M.
- Label the 300 news rows; report news-domain metrics and intra-annotator agreement.
- Fine-tune FinBERT on tweets + news as an optional higher-accuracy tier.
- Systemic impact model on index and yield reactions.
- Longer live capture to evaluate real-time precision of actionable signals.